2-2 Storage

2-2-1 Storage Precautions

When the motor is not in immediate use, take the following storage precautions to ensure that the motor is not exposed to moisture, dust and dirt, or malfunction or damaged due to careless treatment. Climate, storage time and the storage facilities will all affect whether the taken precautions are sufficient. The malfunction or damage caused by improper storage is not covered under warranty, even if the motor is installed on site but has not been put into use. Be sure to observe the following storage precautions and save complete maintenance and repairing records. These records will also affect whether the motor is suitable for powering operation. Components related to assembly, such as fixed surface, parts, especially shaft ends, keys, shaft center holes, etc., should be greased to prevent rust, also prevent damage and deformation caused by impact from external objects and loss its original function.

2-2-2 Storage Environment

- The ideal storage environment should be clean, warm, and well ventilated.
- The ideal ambient temperature should be 10–50°C.
- The motor should be storage in an environment without moisture and not too hot (for example, near a boiler) or too cold (for instance, near a freezer).
- It must be a dry and well-ventilated indoor environment, without direct sunlight, little dust, no corrosive gases (chlorine, sulfur, hydrogen, and its oxides, etc.) and smoke. Make sure there are no pests or insects such as termites, also no flooding is considered.
- You can use a pallet to raise the motor to prevent moisture and ground contamination.
- If the motor has to be stored outdoors for specific reasons, it must be covered by a well waterproof canvas to prevent damage from external objects or other contamination, but good ventilation must still be maintained. Such temporary storage should be kept as short as possible (no more than one month) to reduce functional differences in the future.
- For water-cooling motors or motors with water-cooling bearings, make sure that the water in the pipes has been completed removed without any residue before storage, to avoid rust and corrosion of the pipes or damage due to freezing during long-term storage.

2-2-3 Moisture Protection

When the motor is installed in a humid environment, the temperature difference after shutdown will cause condensation inside the motor. The water vapor condenses on the winding and bearing surfaces will ultimately lead to insulation failure and bearing corrosion. Since the moisture is extremely harmful to electrical components, the motor must be maintained at 3°C above the dewpoint by internal or external heating. The Delta Solution uses the drive DC operation to heat the coil (rotor does not rotate), increase the temperature by 5–10°C above the ambient temperature (W = 2DL / W: wattage, D: stator outer diameter, L: length) and evaporate the internal condensed water to avoid shortening the bearing life. The rated humidity of storage and transportation must not exceed 95%.